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Bank Liquidity Provision across the Firm Size Distribution

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1 Big picture

- **Question.** Why do small firms and large firms get very different access to bank liquidity, especially in a crisis? The paper asks whether the answer lies not just in bank balance-sheet capacity or borrower demand, but in the *terms of the contracts* that govern who can actually draw funds when bad times arrive.
- **Setting.** The paper studies the U.S. corporate loan market using the Federal Reserve Y-14 supervisory loan-level data. The sample covers roughly 60% of all corporate loan commitments and includes more than 50,000 small and medium enterprises, making it much richer than syndicated-loan datasets that mostly observe large firms.
- **Core challenge.** A firm may appear to have liquidity “on paper” through an undrawn credit line, but that line may not be equally usable across firms. Small firms often have contracts that let banks exercise much more discretion, so the same posted credit limit can mean very different true liquidity.
- **Main conceptual contribution.** The paper frames loan design as a trade-off between *lender commitment* and *lender discretion*. Commitment provides insurance against liquidity shocks because the borrower can draw up to an ex-ante limit. Discretion allows the lender to monitor and decide ex post whether to provide funds, which can be valuable when borrower quality is uncertain but makes liquidity less reliable in bad states.
- **Four core facts about loan terms.**
 - Small firms have much shorter-maturity credit lines, including many demand loans that are callable immediately.
 - Small firms almost always post collateral, especially accounts receivable and inventory on credit lines.
 - In normal times small firms use credit lines more heavily, but around distress events they draw less than large firms.
 - Small firms still pay higher spreads even after conditioning on many observable characteristics.
- **Main crisis result.** At the onset of COVID-19, bank credit outstanding surged almost entirely because large firms drew on pre-existing committed credit lines. Small firms, despite having lines outstanding, had essentially no net drawdown in 2020Q1 and then contracted borrowing in 2020Q2.
- **Main heterogeneity result.** Large firms in industries more exposed to the COVID shock drew much more. Small firms in equally exposed industries did not. The difference is strongest early in the crisis and remains after rich controls.
- **Main interpretation.** The paper argues that the difference is not well explained by bond-market substitution, local demand shocks, or weak bank balance sheets. Instead, the evidence points to restrictive loan terms giving banks more discretion over small-firm liquidity.
- **Main policy result.** When the authors match Paycheck Protection Program recipients into the supervisory data, they find that PPP recipients later repaid substantial amounts of non-PPP

bank debt. The interpretation is that government credit largely overcame the private-liquidity shortfall faced by small firms.

- **Economic intuition.** In normal times, discretion can be useful: it lets banks lend to riskier and more opaque borrowers because banks can monitor and pull back if fundamentals look bad. In a large aggregate shock, however, that same discretion means small firms cannot count on their lines the way large firms can.
- **Takeaway.** The paper’s central message is simple but important: credit availability in a crisis is not the same thing as the undrawn commitment written in the contract. For small firms, effective liquidity depends heavily on contract design.

2 Data and measurement (key objects)

2.1 Why the Y-14 data matter

- The main data source is the Federal Reserve FR Y-14Q supervisory collection.
- It covers bank holding companies, IHCs, and savings-and-loan holding companies with more than \$100 billion in consolidated assets during most of the sample.
- The corporate loan schedule contains loan-level information on all commitments of at least \$1 million.
- The key empirical advantage is that the Y-14 observes both large firms and a very large number of SMEs, unlike DealScan or other syndicated-loan sources.

Interpretation. The paper is able to compare loan terms across the firm-size distribution in a way that most U.S. datasets simply cannot.

2.2 Loan coverage and firm-size bins

The authors aggregate loan commitments to the borrowing-entity level and classify firms into five size bins based on assets:

- under \$50 million,
- \$50–\$249 million,
- \$250–\$999 million,
- \$1–\$5 billion,
- above \$5 billion.

They often call firms with assets below \$250 million *SMEs*, and those below \$50 million *small SMEs*. Table 1 shows how large the Y-14 sample is:

- 51,248 firms with assets below \$50 million,
- 11,469 firms with assets between \$50 and \$250 million,

- 4,830 firms with assets between \$250 million and \$1 billion,
- 3,176 firms with assets between \$1 and \$5 billion,
- 2,412 firms above \$5 billion.

Median committed credit rises sharply with firm size: about \$2.6 million for the smallest firms versus about \$44 million for firms above \$5 billion.

2.3 What exactly is in the loan data

For each facility, the Y-14 reports:

- commitment amount,
- utilized amount,
- loan type,
- interest rate,
- origination and maturity dates,
- renewal flags,
- whether the loan is secured,
- the collateral type,
- internal bank risk ratings,
- borrower industry, location, assets, and some financial statement information.

The main loan categories are corporate C&I loans, owner-occupied nonfarm nonresidential real-estate loans, agricultural production loans, and other leases.

2.4 Coverage relative to the universe of loans

The paper benchmarks Y-14 against the Y-9C universe. At the end of 2019:

- Y-9C reported \$4.61 trillion in corporate commitments and \$2.25 trillion in corporate loans outstanding.
- The 29 large Y-14 banks in the sample had \$3.54 trillion in Y-9C commitments.
- The Y-14 schedule itself captures \$2.77 trillion of corporate commitments, about 60% of total Y-9C lending.

The paper also emphasizes two limitations:

- it excludes commitments below \$1 million,
- and it excludes lending by smaller banks outside the Y-14 reporting population.

Interpretation. The data are not the full universe, but they are broad enough to make the paper's cross-size comparisons macroeconomically meaningful.

2.5 Key variables used throughout the paper

Maturity and remaining maturity.

The paper studies both the maturity at origination or renewal and the maturity remaining as of late 2019. This distinction is important because large firms actively renew long-maturity credit lines well before expiration, while small firms typically do not.

Collateral.

Collateral is coded into categories such as real estate, fixed assets, accounts receivable and inventory (AR&I), cash, other assets, blanket lien, and unsecured.

Utilization / drawdown.

The drawdown rate is the ratio of utilized to committed credit. This is the key dependent variable in the COVID section.

Exposure to the COVID shock.

The main exposure measure is the abnormal decline in national employment in the firm's three-digit NAICS industry between 2019Q2 and 2020Q2, relative to the pre-COVID trend. A retail-sales-based exposure measure is used as a robustness check.

PPP matching.

The authors hand-match the Small Business Administration's PPP recipient file into the Y-14 borrower records using names and addresses.

2.6 Descriptive facts that motivate the theory

Table 2 shows an important background pattern: small firms are materially more opaque.

- The fraction of loans to firms with any audited financial statement ever reported rises from 27% for firms below \$50 million in assets to 90% for firms above \$5 billion.
- The fraction with an audit date in the last two quarters rises from 6.5% to 41%.
- The average lag since the latest audit date falls from 4.8 quarters to 3.1 quarters.

The paper also shows in the appendix that smaller firms have more volatile revenues, earnings, and stock returns.

Interpretation. These are exactly the kinds of differences that make lender discretion more attractive in the model and more likely in small-firm lending in the data.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism: commitment versus discretion

The theoretical idea is that a bank credit line can be designed to provide either:

- **commitment:** the borrower can draw up to a fixed limit decided at date 0, or
- **discretion:** the lender can learn more at date 1 and decide whether to provide funds.

The trade-off is straightforward:

- commitment provides liquidity insurance,
- discretion allows monitoring and protects the lender when borrower quality is uncertain.

3.2 Model setup

A firm has assets of value A and faces a date-1 cash-flow shock ρ per unit of assets. If it cannot inject funds of size ρ , the firm fails. At date 2 each unit of assets yields terminal payoff

$$z + \Delta, \tag{1}$$

where Δ is mean-zero uncertainty in long-run value.

The key financing friction is limited pledgeability: the firm can promise only a fraction θ of terminal value to the lender.

Committed credit.

If the lender commits ex ante to a credit limit $\hat{\rho}$, the largest feasible committed line solves the break-even condition

$$\int_{-\infty}^{\hat{\rho}} (\theta z - \rho) dF(\rho) = 0. \tag{2}$$

Under normality, the paper shows this implies

$$\hat{\rho} = \mu + \sigma h^{-1}\left(\frac{\mu - \theta z}{\sigma}\right), \quad h(x) = \frac{\phi(x)}{\Phi(x)}, \tag{3}$$

and importantly $\hat{\rho} > \theta z$.

Interpretation. Commitment lets good states cross-subsidize bad states. Ex post the lender would prefer to liquidate once $\rho > \theta z$, but ex ante commitment supports a higher limit because the lender averages across states.

Discretionary credit.

If the contract gives the lender discretion, the lender can pay monitoring cost ζ at date 1, observe more about Δ , and then decide whether to lend.

Without monitoring, lender value is

$$V^N(\rho) = \max\{\theta z - \rho, 0\}. \tag{4}$$

With monitoring, lender value is

$$V^M(\rho) = \mathbb{E}[\max\{\theta(z + \Delta) - \rho, 0\}] - \zeta. \tag{5}$$

The lender monitors when $V^M(\rho) > V^N(\rho)$, which happens for an intermediate range of cash-flow shocks.

Interpretation. Small shocks are not alarming enough to justify monitoring. Very large shocks are too bad to save. Monitoring is most valuable in the middle region.

3.3 When discretion is preferred

The paper shows two necessary conditions for discretion to dominate commitment:

- uncertainty over terminal repayment ability must be high enough,
- expected cash-flow needs must be large relative to pledgeable value, formally $E[\rho] > \theta z$.

This maps naturally to small firms because they tend to have:

- less reliable and less frequently audited financials,
- more volatile sales and earnings,
- lower pledgeability and more opaque balance sheets.

Prediction. Small firms should receive loan terms that give lenders more discretion: short maturity, demandable lines, frequent rollover, and collateral structures that can be marked to market.

3.4 Main empirical drawdown regression

The central COVID regression relates credit-line drawdowns to firm size interacted with the COVID period:

$$\text{Drawdown}_{f,i,t} = \alpha + \delta_t + \sum_{s \neq \text{small SME}} \beta_s \mathbf{1}\{\text{size}_i = s\} \times \text{COVID}_t + \varepsilon_{f,i,t}. \quad (6)$$

Here:

- $\text{Drawdown}_{f,i,t}$ is utilization divided by commitment for facility f of firm i ,
- COVID_t indicates 2020Q1 or 2020Q2,
- the omitted group is firms below \$50 million in assets.

The richer specifications add:

- loan fixed effects,
- bank-time fixed effects,
- state-time and industry-time fixed effects,
- lagged firm financials interacted with COVID,
- internal rating-time fixed effects,
- loan-term controls (maturity, collateral, spreads, pre-COVID utilization) interacted with COVID.

Interpretation. The coefficient β_s measures how much more a larger size class drew during COVID relative to small SMEs, holding constant increasingly rich sets of demand and supply controls.

3.5 Exposure-based triple-difference specification

To ask whether firms in equally exposed industries behaved differently by size, the paper estimates:

$$\begin{aligned} \text{Drawdown}_{f,i,t} = & \alpha + \delta_t + \sum_s \beta_{1,s} \mathbf{1}\{\text{size}_i = s\} \times \text{COVID}_t \\ & + \beta_2 \text{Exposure}_i \times \text{COVID}_t \\ & + \sum_s \beta_{3,s} \text{Exposure}_i \times \mathbf{1}\{\text{size}_i = s\} \times \text{COVID}_t + \varepsilon_{f,i,t}. \end{aligned} \quad (7)$$

Interpretation.

- β_2 is the exposure effect for the omitted size class (small SMEs).
- The $\beta_{3,s}$ terms show how much more sensitive larger firms are to COVID exposure than small SMEs.
- If credit contracts are truly more restrictive for small firms, then exposure should translate into much larger drawdowns for large firms than for small firms.

3.6 Instrumental-variable specification

The employment-based exposure measure is instrumented using physical proximity requirements in the industry, derived from O*NET.

Conceptually, the IV equation is:

$$\Delta \text{Drawdown}_{i,2020Q1-2019Q4} = \beta \text{Exposure}_i + u_i, \quad (8)$$

with Exposure_i instrumented by industry-level physical proximity.

Interpretation. Industries requiring close physical interaction were hit harder by pandemic restrictions for reasons plausibly orthogonal to pre-existing credit-line design. This strengthens the causal interpretation of the size gradient in exposure sensitivity.

3.7 Pricing regression

To study whether small firms pay more even after conditioning on observables, the paper estimates loan-pricing regressions of the form:

$$\text{Interest}_{f,i,b,t} = \sum_s \gamma_s \mathbf{1}\{\text{size}_i = s\} + \Gamma X_{f,i,b,t} + \eta_{f,i,b,t}, \quad (9)$$

where $X_{f,i,b,t}$ progressively adds industry-time, bank-time, rating-time, firm financial, maturity, collateral, and line-size controls.

Interpretation. If small firms still pay higher spreads after these controls, then loan terms are not just substituting for price. Instead, small firms appear to select into a different lending regime with stricter nonprice terms *and* higher prices.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimates

There are two distinct identification layers in the paper.

First, the cross-sectional loan-term facts are identified from rich differences across firm size classes inside the same supervisory dataset.

Second, the crisis analysis identifies the differential access to liquidity from within-facility changes in utilization during COVID, using loan fixed effects and then adding bank-time, state-time, and industry-time controls.

Interpretation. The COVID regressions are not simply comparing different firms at one date; they compare the same lines before and during the crisis, while asking whether size predicts a different response.

4.2 Why reverse causality is weak

The contract terms at the center of the story — maturity, demandability, collateral structure — are largely predetermined before the COVID shock. Small firms did not become short-maturity borrowers because they failed to draw in 2020Q1; rather, they entered the shock already holding these restrictive contracts.

Interpretation. This timing is crucial. The paper is not estimating a simultaneous system in which drawdowns and contract terms are chosen together in real time during the pandemic.

4.3 How the paper addresses demand-side confounds

A major concern is that large firms might simply have had larger liquidity needs. The paper addresses this in several ways:

- it controls for industry and state shocks,
- it controls for balance-sheet characteristics such as leverage, profitability, cash, sales growth, and internal ratings,
- it controls for bond-market access, since some large firms may have used lines because public debt markets were disrupted in March 2020,
- and it validates the exposure measure by showing that industry exposure predicts actual revenue declines similarly for small and large firms.

Table 12 is especially important here: a one-standard-deviation increase in exposure is associated with about a 15 percentage point decline in revenue growth for both small and large firms. So the paper shows similar real shocks but very different drawdown behavior.

4.4 Why the bond-market explanation is incomplete

One alternative story is that large firms drew heavily because they were temporarily shut out of bond markets. The paper directly addresses this:

- it adds a bond-market-access interaction,
- it shows that the size gradient remains large,
- and Table 10 shows strong drawdowns even among firms that never accessed public bond markets.

Interpretation. Bond-market substitution may explain part of the aggregate surge in large-firm drawdowns, but it does not explain why small firms in equally exposed industries did not draw.

4.5 Why the paper emphasizes incomplete contracts rather than simple supply versus demand

The authors explicitly argue that neither a pure credit-demand story nor a pure bank-balance-sheet story is sufficient.

- Demand alone does not explain why firms facing similar industry-level shocks respond so differently by size.
- Bank balance-sheet weakness alone does not explain the results because bank-time fixed effects absorb supply differences across banks and bank capital or liquidity do not explain the drawdown gradient.

Interpretation. The paper’s contribution is to shift the discussion from “who wanted funds” or “which banks were weak” to “who had contracts that actually allowed access to funds in bad states.”

4.6 Bank balance sheets versus economic environment

The paper provides direct evidence against the view that weak banks were the main driver.

- Bank-time fixed effects leave the size gradient largely intact.
- Appendix evidence shows that capital, liquid assets, and deposits do not explain the 2020Q1 drawdown gradient.
- The Senior Loan Officer Survey indicates that banks tightened mainly because of worsening economic outlook, not because of capital or liquidity problems.

Interpretation. The more convincing story is that a sharp deterioration in borrower outlook interacted with pre-existing discretion in small-firm contracts.

4.7 Role of the instrument

Physical proximity requirements are used as an excluded instrument for the abnormal employment decline in 2020. This helps because actual employment declines could still embed endogenous industry responses or omitted industry shocks.

In Table 14 the instrument is strong, with effective first-stage F -statistics generally near or above conventional relevance thresholds. The IV results preserve the main qualitative pattern: essentially no response for the smallest firms and much larger effects for the biggest firms.

4.8 What the paper can and cannot fully pin down

The authors are careful about several limitations:

- The model is illustrative rather than a full structural estimation.
- The Y-14 does not report covenant tightness or MAC clauses, so some dimensions of lender discretion are unobserved.
- Loan terms such as maturity and collateral are endogenous, so the reduction in coefficients after controlling for them is suggestive rather than a fully causal decomposition.
- The sample excludes sub-\$1 million loans and lending by smaller banks.
- The PPP matching may miss some recipients, and Y-14 does not include PPP loans themselves.

Interpretation. These limitations do not overturn the paper’s argument, but they matter for how precisely one should interpret the size of each channel.

4.9 Relation to the literature

The paper contributes to several literatures at once:

- loan contract design and incomplete contracting,
- credit lines as liquidity insurance,
- SME finance and relationship banking,
- bank lending in the COVID recession,
- and public credit support through PPP.

Its distinctive point is that small firms’ crisis liquidity constraints were not mainly about lacking lines in the first place. They were about holding lines with much more lender discretion.

5 Tables and figures

5.1 Table 1 and Table 2: the data and the underlying firm differences

Read:

- Table 1 shows the sample is large enough to meaningfully compare small and large firms. The smallest size bin alone contains more than 51,000 firms, with median total commitment of only \$2.6 million. The largest size bin contains about 2,400 firms, with median commitment of \$44 million.

Table 1
Distribution of committed bank credit by firm type and firm size.

Firm size (Assets in millions)	Committed credit (in \$mil.)						Firms in category
	1st Percentile	10th Percentile	Mean	Median	90th Percentile	99th Percentile	
Panel A: All Firms							
Unclassified	1.0	1.1	15.2	2.3	15.1	225.0	24,824
0-50	1.0	1.1	5.6	2.6	13.5	37.8	51,248
50-250	1.0	1.8	30.6	14.6	74.6	220.5	11,469
250-1000	1.0	2.0	99.8	25.6	253.8	938.0	4830
1000-5000	1.0	2.4	300.4	43.9	894.8	2,835.0	3176
5000+	1.0	2.2	612.4	44.0	1,861.5	6,607.5	2412
Panel B: CompuStat							
0-50	1.0	1.0	5.6	2.7	13.3	32.2	1004
50-250	1.0	1.6	41.7	20.0	100.0	333.5	434
250-1000	1.0	1.8	134.8	48.9	367.5	1,196.0	707
1000-5000	1.0	3.5	436.1	118.3	1,272.9	3,077.1	1145
5000+	1.2	4.7	981.4	215.5	2,918.0	7,611.8	1109
Panel C: Syndicated Bank Loans							
0-50	1.5	3.7	28.0	11.1	56.1	264.6	202
50-250	2.0	7.2	68.7	30.0	133.0	460.6	652
250-1000	2.9	11.2	140.7	93.1	375.0	783.6	588
1000-5000	4.0	20.6	381.9	224.9	863.8	2,313.3	911
5000+	6.0	78.3	1,071.7	650.1	2,762.3	6,000.0	520

Notes: The table reports the distribution of firm-level committed credit by firm size group. Firm-level commitments are constructed by summing over credits in the Y-14 data. For syndicated credits, the reported participation interest is scaled up to reflect the total commitment, and loans held by multiple Y-14 banks are de-duplicated. The sample includes all CFI loans to U.S. addresses, corporate loans secured by owner-occupied nonfarm nonresidential properties, loans to finance agricultural production, and other leases. Panels B and C restrict to firms that appear in CompuStat or have syndicated loans, respectively.

Table 2
Frequency of borrower financials.

Assets (mil.)	Financials date			Audit date			Obs.
	Ever	Last 2Q	Lag (Qtrs.)	Ever	Last 2Q	Lag (Qtrs.)	
0-50	0.96	0.4	3.2	0.27	0.065	4.8	622,257
50-250	0.96	0.48	2.9	0.68	0.17	4.3	212,128
250-1000	0.93	0.47	2.9	0.82	0.25	3.9	146,600
1000-5000	0.93	0.53	2.6	0.88	0.35	3.4	170,367
5000+	0.93	0.59	2.5	0.9	0.41	3.1	163,265

Notes: The table summarizes the frequency with which the date of financials (or audited financials) is ever reported, whether there is a reported date in the last 2Q, and the average time since the reported date (in quarters) conditional on a date being reported. Sample is 2015Q1-2019Q4. Excludes bank-quarters that rarely report audit dates. Observation count reports the total number of loan-quarters in each size category, regardless of financials reporting.

- Table 2 shows a sharp size gradient in transparency. Only 27% of loan-quarters for the smallest firms ever report an audited financial statement, compared with 90% for the largest firms.

Economic message. The paper is not just documenting that small firms are smaller. It is documenting that they are also more opaque and harder to verify, which is exactly the kind of environment in which lender discretion becomes valuable.

5.2 Figure 1 and Tables 3–6: maturity, rollover, and collateral as control rights

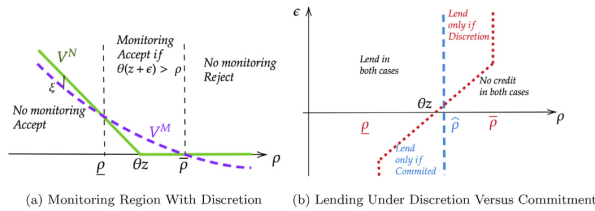


Fig. 1. Model properties.

Table 3
Maturity at origination/renewal by facility type and firm size category as of December 31, 2019.

Maturity at origination/renewal	Demand	<1 year	1 year	1-2 year	2-4 years	4-5 years	>5 years	Obs.
Assets (\$mil.)								
Panel A: Revolving Credit Lines								
0-50	0.29	0.23	0.23	0.16	0.058	0.028	0.013	26,924
50-250	0.15	0.12	0.1	0.15	0.19	0.28	0.03	8089
250-1000	0.076	0.046	0.04	0.066	0.17	0.56	0.045	5924
1000-5000	0.024	0.021	0.021	0.033	0.15	0.71	0.047	6598
5000+	0.018	0.039	0.059	0.042	0.12	0.67	0.048	6199
Panel B: Term Loans								
0-50	0.0012	0.041	0.022	0.015	0.07	0.26	0.59	13,612
50-250	0.0013	0.04	0.022	0.024	0.14	0.43	0.34	6222
250-1000	0.00061	0.032	0.014	0.034	0.13	0.48	0.31	3293
1000-5000	0	0.037	0.017	0.033	0.16	0.53	0.22	2587
5000+	0.0005	0.071	0.048	0.087	0.25	0.38	0.16	1982

Notes: The table reports the fraction of outstanding loans to each firm size group (assets in \$million) by the maturity indicated in the table header. The maturity is as of the respective facility's origination date or alternatively the most recent renewal date if the facility has been renewed since origination. The sample includes loans as of December 31, 2019 for which an origination or renewal date is reported.

Read:

- Figure 1 visualizes the model. Under discretion, monitoring is only worthwhile for intermediate shocks, and lending can be denied even when the borrower is ex post worth saving.
- Table 3 shows that 29% of revolving lines to firms below \$50 million are demand loans, and roughly three-quarters have original maturity of one year or less. For firms above \$1 billion, the modal revolving line is a five-year facility and less than 10% originate below two years.
- Table 4 shows that small firms do almost no active maturity management: one-year lines are renewed right at expiration. Large firms renew medium- and long-maturity lines much earlier.
- Table 5 makes the remaining-maturity result very stark: about 40% of the smallest firms' lines were callable or due already in 2020Q1, and about 85% were due sometime in 2020. For the largest firms, only about 15% had less than one year remaining.
- Table 6 shows that small firms almost always pledge collateral. Among non-demand revolving credit lines, less than 10% of small-SME lines are unsecured, versus 71% for the largest firms.

For term loans, only 2% of loans to the smallest firms are unsecured, versus 44% for the largest firms.

Table 4
Maturity management in revolving credit lines and term loans by firm size category

Assets (\$mil.)	1 year or less				1-2 years				2-4 years				More than 4			
	Before		After		Before		After		Before		After		Before		After	
	N		N		N		N		N		N		N		N	
Panel A: Credit Lines																
0-50	0	12	274076	1	19	7318	6	31	29077	56	61	17479				
50-250	0	12	40580	6	25	3026	12	34	38100	38	60	44375				
250-1000	0	12	12813	9	22	10051	1	25	34075	39	60	48390				
1000-5000	0	12	7625	11	19	7188	26	36	43873	38	60	106256				
5000+	1	12	14996	12	20	7116	28	36	36800	44	60	136449				
Panel B: Term Loans																
0-50	0	4	13630	2	14	8975	19	15	30922	47	68	142379				
50-250	0	6	8054	6	16	5577	21	13	29443	42	60	93464				
250-1000	0	9	3028	12	14	2654	25	13	16214	43	59	50240				
1000-5000	1	11	2037	10	20	2142	20	13	14880	45	59	41247				
5000+	1	7	5221	12	14	3893	29	34	14902	68	59	27493				

Notes: The table reports the median maturity (in months) before and after a credit facility is renewed. Facilities are grouped by their maturity at origination/renewal (normal date as noted in the header). Demand loans are excluded from the sample. The sample is restricted to all intervals of revolving credit lines (Panel A) and term loans (Panel B) reported between 2015Q1 through 2019Q4.

Table 5
Renewing maturity by facility type and firm size category for loans outstanding on December 31, 2019.

Loan size	Demand	Jan	Feb	Mar	Q2	Q3-Q4	2021	2022-24	Last	Obs.
Panel A: Revolving Credit Lines										
0-50	0.28	0.029	0.029	0.032	0.19	0.27	0.1	0.055	0.0889	27.067
50-250	0.19	0.016	0.016	0.018	0.041	0.15	0.17	0.33	0.031	14.061
250-1000	0.11	0.0034	0.0038	0.0039	0.039	0.074	0.13	0.39	0.035	8.142
1000-5000	0.096	0.0021	0.0028	0.0017	0.018	0.041	0.1	0.71	0.0099	6.662
5000+	0.078	0.0009	0.0009	0.0008	0.022	0.053	0.082	0.72	0.014	6.662
Panel B: Term Loans										
0-50	0.0015	0.0043	0.0056	0.0063	0.018	0.036	0.063	0.36	0.5	22.541
50-250	0.0015	0.0017	0.0004	0.0005	0.0075	0.02	0.042	0.12	0.35	0.24
250-1000	0.0011	0.0023	0.0004	0.0002	0.019	0.041	0.11	0.62	0.2	4.087
1000-5000	0	0.0054	0.0027	0.0027	0.019	0.04	0.097	0.68	0.14	3.333
5000+	0.0008	0.014	0.011	0.01	0.04	0.085	0.14	0.38	0.12	2.988

Notes: The table reports the fractions of loans outstanding in each size group (columns in brackets) in the table below. The sample includes loans from October of December 31, 2019.

Notes: The table reports the fraction of loans to each firm size group (assets in \$milions) with remaining maturity indicated in the table header. The sample includes loans outstanding as of December 31, 2019.

Table 6
Collateral use by facility type and firm size category as of December 31, 2019.

Collateral type	Real estate	Cash	AR & inventory	Fixed assets	Other	Blanket lien	Unsecured	Obs.
Assets (\$mil.)								
Panel A1: Revolving Credit Lines (Non-Demand Loans)								
0-50	0.025	0.015	0.47	0.029	0.049	0.38	0.037	25762
50-250	0.025	0.026	0.44	0.037	0.083	0.28	0.082	9792
250-1000	0.015	0.043	0.37	0.048	0.11	0.25	0.17	7073
1000-5000	0.0064	0.039	0.31	0.039	0.11	0.18	0.32	6986
5000+	0.0019	0.034	0.1	0.015	0.074	0.074	0.71	7987
Panel A2: Revolving Credit Lines (Demand Loans)								
0-50	0.0677	0.012	0.66	0.034	0.049	0.16	0.079	10542
50-250	0.0093	0.026	0.37	0.044	0.037	0.11	0.37	3901
250-1000	0.0017	0.02	0.18	0.049	0.019	0.058	0.85	1771
1000-5000	0.0007	0.022	0.11	0.0056	0.012	0.046	0.83	1423
5000+	0	0.015	0.053	0.041	0.02	0.026	0.88	884
Panel B: Term Loans								
0-50	0.48	0.0044	0.11	0.12	0.025	0.25	0.019	22008
50-250	0.34	0.012	0.11	0.11	0.043	0.23	0.006	8817
250-1000	0.14	0.029	0.13	0.19	0.056	0.24	0.003	6382
1000-5000	0.074	0.028	0.14	0.18	0.086	0.23	0.006	5313
5000+	0.02	0.018	0.082	0.23	0.068	0.15	0.44	2397

Source: Compustat. The table reports the average size of loans in each collateral category for firms in the sample (the assets in \$millions) for the years 1990-2013. The table reports the average size of loans in each collateral category for firms in the sample (the assets in \$millions) for the years 1990-2013. The table reports the average size of loans in each collateral category for firms in the sample (the assets in \$millions) for the years 1990-2013. The table reports the average size of loans in each collateral category for firms in the sample (the assets in \$millions) for the years 1990-2013. The table reports the average size of loans in each collateral category for firms in the sample (the assets in \$millions) for the years 1990-2013. The table reports the average size of loans in each collateral category for firms in the sample (the assets in \$millions) for the years 1990-2013. 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Notes: The table reports the fraction of loan commitments to each firm size group (by assets in \$milions) with the type of collateral indicated in the table header. The sample includes loans as of December 31, 2019.

Economic message. Short maturity and collateral are not just “contract features.” In the paper’s interpretation they are effective control rights. Banks retain frequent opportunities to refuse rollover or to tighten lending when small firms want to draw.

5.3 Figure 2 and Tables 7–9: utilization in normal times and pricing

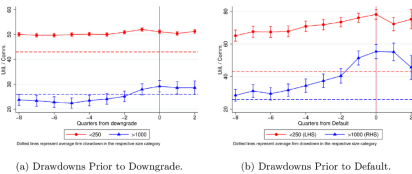


Fig. 2. Drawdowns prior to downgrade/default by firm size. Data are restricted to a balanced panel of loans where utilization is observed for all eight quarters prior to and including downgrade or default. The balanced panel restriction is not applied to quarters after the downgrade or default occurs, and those quarters are depicted in this graph for illustrative purposes. Downgrade is measured at the loan-level. Default is defined as a firm being placed on non-accrual. Confidence intervals for the mean drawdown by firm size based on robust standard errors and were estimated as in Cattaneo et al. (2019).

Table 7
Drawdown of revolving credit lines by firm size, 2019Q4.

Assets (mil.)	Utilization/commitment								Δ Util/ Comm. %	Obs.
	< 10%	10-30%	30-50%	50-70%	70-90%	> 90%	Δ Util/ Comm. %	Obs.		
0-50	0.33	0.087	0.12	0.15	0.14	0.17	10	36,827		
50-250	0.35	0.1	0.12	0.15	0.14	0.14	10	10,928		
250-1000	0.37	0.12	0.14	0.15	0.12	0.094	8.4	8122		
1000-5000	0.47	0.16	0.13	0.11	0.075	0.067	7.7	9447		
>5000	0.77	0.08	0.053	0.03	0.014	0.056	4.5	8729		

Notes: The table reports the distribution of drawn credit as share of total commitments and the average change in the absolute value of drawn credit as a share of total commitments. The distribution is reported for 2019Q4. Changes in drawn credit are based on the period 2015Q1 through 2019Q4. Observations report the number of loans in each size category in 2019Q4.

Read:

- Table 7 shows that in late 2019 small firms used their lines much more heavily than large firms. Nearly one-third of the smallest SMEs had utilization above 70%, while only about 7% of the largest firms did. At the same time, three-quarters of the largest firms had utilization below 10%.
- Figure 2 shows that around downgrade and default events, large firms ramp up utilization before trouble, while small firms barely do. This is one of the paper’s clearest pieces of evidence that discretion matters precisely when conditions worsen.
- Table 8 shows the full distribution of rates shifts downward with firm size: larger firms systematically face lower interest rates.
- Table 9 shows the spread gradient survives rich controls. Large firms pay more than 100 basis points less than the smallest SMEs unconditionally, and still around 80 basis points less after controlling for industry, lender, ratings, and firm financials. Adding maturity and collateral shrinks the gap further, but does not erase it.

Economic message. Small firms get stricter nonprice terms *and* still pay more. That pattern is hard to square with the idea that they merely choose different points on the same pricing menu. It fits much better with the idea that they are fundamentally different borrowers in a high-discretion lending regime.

Table 8
Interest rates by facility type and firm size category on December 31, 2019.

Interest in bp	0-100	100-200	200-300	300-400	400-500	500-600	> 600	Obs.
Assets (\$mil.)								
Panel A: Revolving Credit Lines								
0-50	0.015	0.01	0.054	0.3	0.41	0.17	0.04	24293
50-250	0.048	0.03	0.16	0.4	0.2	0.076	0.083	7392
250-1000	0.068	0.026	0.16	0.34	0.22	0.091	0.1	5489
1000-5000	0.086	0.017	0.21	0.37	0.16	0.078	0.075	5817
5000~	0.2	0.02	0.24	0.32	0.11	0.053	0.053	2623
Panel B: Term Loans								
0-50	0.015	0.0018	0.029	0.38	0.42	0.13	0.026	22541
50-250	0.024	0.0031	0.074	0.49	0.28	0.079	0.054	8826
250-1000	0.026	0.0059	0.11	0.47	0.24	0.076	0.07	4386
1000-5000	0.035	0.011	0.2	0.54	0.13	0.045	0.036	3333
5000~	0.094	0.019	0.26	0.46	0.12	0.029	0.013	2598

Notes: The table reports the fraction of loan commitments to each firm size group (by assets in \$million) with the interest rate indicated in the table header. Interest rates represent the reference rate plus spread for floating rate loans and fixed interest rate for fixed rate loans, both as of December 31, 2019. Interest rates for revolving credit lines are only reported if the drawdown is strictly larger than zero. The sample includes loans as of December 31, 2019.

Table 9
Pricing of revolving credit lines and term loans by firm size category.

Dependent variable	Interest rate (in bp)						
	Credit lines			Term loans			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
50-250 (in mil)	-62.0*** (2.1)	-36.3*** (1.7)	-35.6*** (1.7)	-35.7*** (1.7)	-17.4*** (2.3)	-12.2*** (1.8)	-11.2*** (1.8)
250-1000	-55.6*** (3.7)	-37.0*** (2.9)	-35.7*** (3.2)	-35.6*** (3.2)	-13.0** (4.1)	-8.7** (3.0)	-5.5 (3.0)
1000-5000	-69.2*** (3.2)	-63.0*** (2.7)	-58.5*** (3.3)	-58.2*** (3.3)	-66.5*** (3.7)	-53.1*** (3.1)	-39.7*** (3.3)
5000~	-113.9*** (4.1)	-85.3*** (4.5)	-76.2*** (5.1)	-76.0*** (5.1)	-107.3*** (4.0)	-79.7*** (3.6)	-63.4*** (3.8)
Reference-Rate-Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Time FE	No	Yes	Yes	Yes	No	Yes	Yes
Bank-Time FE	No	Yes	Yes	Yes	No	Yes	Yes
Rating-Time FE	No	Yes	Yes	Yes	No	Yes	Yes
Firm Financial Controls	No	Yes	Yes	Yes	No	Yes	Yes
Loan Terms Controls	No	No	Yes	Yes	No	No	Yes
Drawdown	No	No	No	Yes	No	No	Yes
No of Firms	41,645	37,172	37,053	37,053	31,208	26,314	26,214
N	130,277	114,102	112,545	112,545	61,320	53,822	52,412
R ²	0.359	0.553	0.566	0.566	0.279	0.521	0.535

Notes: Results from estimating a model of the following type: $\text{Interest}_{i,t} = \sum_{b \in \{50, 250, 1000, 5000\}} \beta_{1,b} [\text{size class} = b] + \Gamma' X_i + \epsilon_{i,t}$ where $\text{Interest}_{i,t}$ is the interest on facility i from bank b to firm i at time t . The sample contains originations and renewals between 2015Q1 and 2019Q4. Industry-time fixed effects are at the NAICS 3-digit level. Rating-time fixed effects are categorical variables for 10 internal loan rating categories. Firm financial controls are lagged debt/assets, cash and receivables/assets, net income/assets, and operating income/interest expense. Loan term controls are six maturity categories (demand loans, 0-6 months, 6-12 months, 1-2 years, 2-4 years, more than 4 years), six collateral classes (real estate, marketable securities, accounts receivables and inventory, fixed assets, other, and unsecured or blanket lien), and total credit line commitment over total assets. Robust standard errors are clustered at the firm level in parentheses; *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

5.4 Tables 10 and 11: the basic COVID drawdown fact

Read:

- Table 10 shows the headline fact. Small SMEs had virtually no increase in total Y-14 credit in 2020Q1 (\$186.3B to \$188.6B), while firms with assets above \$1 billion experienced a combined increase of roughly 44%.
- The same table shows that the action was in *drawdowns on pre-existing credit lines*, not mainly in term loans or new lending.
- Table 11 translates this into regression estimates. In the simplest specification, firms above \$5 billion draw about 14.1 percentage points more than small SMEs in COVID quarters. After adding bank-time, state-time, industry-time, ratings, and financial controls, the gap is still about 9.2 percentage points.
- When the authors add loan-term controls, the large-firm advantage drops materially, to roughly 5.3 percentage points in the richest specification.

Table 10
Aggregate drawdowns in \$B by firm type, 2019Q4-2020Q2.

	Total Y-14 credit			Term loans			CL drawdowns (all facilities)			CL drawdowns (pre-existing facilities)		
	2019Q4	2020Q1	2020Q2	2019Q4	2020Q1	2020Q2	2019Q4	2020Q1	2020Q2	2019Q4	2020Q1	2020Q2
Panel A: By Firm Size (in Assets in \$mil)												
Not classified	138.0	141.5	141.1	58.7	60.5	62.4	48.2	51.5	46.2	44.4	48.0	39.4
0-50	186.3	188.6	159.7	67.5	67.1	67.1	102.4	104.4	73.5	99.7	102.0	70.9
50-250	187.1	193.8	169.0	62.2	62.5	59.4	102.2	109.9	86.0	100.8	106.9	83.6
250-1000	185.2	212.7	186.5	56.9	57.2	53.1	105.8	133.1	109.6	103.4	131.4	107.3
1000-5000	238.6	317.6	266.5	77.4	82.2	77.9	125.3	199.0	151.4	124.1	197.8	149.0
5000~	240.2	373.4	300.1	97.9	118.2	113.3	73.6	184.6	115.2	72.2	182.7	111.8
Sum	1175.3	1427.6	1222.8	420.6	448.3	433.1	557.5	781.5	581.9	544.7	768.6	562.0
Panel B: Other Firm Characteristics												
Bond Market Access	332.9	503.7	407.0	125.2	146.4	139.3	126.6	277.1	185.5	127.6	275.0	181.5
Bond Issued March-July	95.5	109.2	124.4	36.8	45.2	39.5	28.0	92.6	54.8	27.7	92.2	54.3
CP facilities	3.2	10.1	5.4	1.1	1.7	1.6	1.8	8.1	3.2	1.8	8.1	3.0

Notes: The table reports the total dollar amount (in \$billions) of utilized credit pooling all facilities (leftmost columns), term loans (second set of columns), revolving credit lines only (third set of columns), and revolving credit lines of firms that had a facility open as of the previous quarter (rightmost columns). The columns headed "Total Y-14 Credit" include non-revolving credit lines, capitalized lease obligations, and other unclassified loan types in addition to term loans and credit line drawdowns. In Panel B, we restrict the sample to firms that have bond market access (the firm either had a bond outstanding according to Compustat-Capital IQ in 2019Q4 or issued a bond at some point from 2010 through 2020 according to Mergent FISD), firms that issued a bond in March-July 2020, and loans with the purpose to back up a commercial paper (CP) facility.

Table 11
Drawdowns by firm size.

Dependent variable	Drawdown rate (in ppt)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
50-250 (in mil) × COVID	4.1*** (0.7)	4.0*** (0.7)	3.0*** (0.7)	2.2*** (0.7)	2.0*** (0.7)	4.1*** (0.8)	0.4 (0.4)	0.5** (0.2)	
250-1000 × COVID	10.5*** (1.2)	10.3*** (1.2)	8.8*** (1.0)	8.6*** (1.0)	6.7*** (1.1)	6.9*** (1.1)	3.9*** (0.9)	3.6*** (0.6)	
1000-5000 × COVID	13.5*** (1.7)	12.6*** (1.6)	10.8*** (1.1)	10.6*** (1.1)	8.8*** (1.0)	9.1*** (1.1)	10.7*** (1.3)	5.3*** (0.7)	4.8*** (0.7)
5000- × COVID	14.1*** (2.4)	12.6*** (2.1)	10.2*** (1.5)	9.9*** (1.5)	8.2*** (1.4)	8.2*** (1.5)	11.4*** (1.7)	5.3*** (1.5)	3.8*** (1.0)
Bond Market × COVID		1.9* (1.0)	1.6 (1.0)	1.6 (0.9)	1.3 (0.8)	1.2 (0.8)	0.3 (0.8)	0.7 (0.8)	-0.0 (0.7)
Loan FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	No	No	No	No	No	No	No
Bank-Time FE	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Time FE	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Time FE	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Financials	No	No	No	No	No	Yes	Yes	Yes	Yes
Rating-Time FE	No	No	No	No	No	Yes	Yes	Yes	Yes
Maturity Controls	No	No	No	No	No	No	No	No	Yes
Collateral Controls	No	No	No	No	No	No	No	No	Yes
Interest Rate Controls	No	No	No	No	No	No	No	No	Yes
Drawdowns in 2019q4	No	No	No	No	No	No	No	No	Yes
No of Firms	62,615	62,615	62,615	62,615	62,614	56,568	15,850	56,568	41,292
N	786,188	786,188	786,188	786,186	786,156	712,177	348,522	712,113	518,256
R ²	0.83	0.83	0.83	0.83	0.83	0.82	0.84	0.86	

Notes: Results from estimating a model of the following type: $\text{Drawdown}_{i,t} = \alpha_i + \beta \times \sum_{b \in \{50, 250, 1000, 5000\}} [\text{size class} = b] \times \text{COVID} + \Gamma' X_i + \epsilon_{i,t}$ where $\text{Drawdown}_{i,t}$ is the ratio of utilized to committed credit and COVID is an indicator for 2020Q1 and 2020Q2. We restrict the sample to outstanding loans from 2019Q4 onwards. Bond Market, indicates whether firm i has issued bonds at any point between 2010 and 2020Q2. Industry-time fixed effects are at the NAICS three-digit level. Rating-time fixed effects are categorical variables for 10 internal loan rating categories. Firm financial controls are lagged debt/assets, cash and receivables/assets, operating income plus depreciation and amortization/assets, year-over-year sales growth, and operating income/interest expense, each interacted with COVID. Maturity and collateral controls are six maturity categories (demand loans, 0-6 months, 6-12 months, 1-2 years, 2-4 years, and more than 4 years) and six collateral classes (real estate, marketable securities, accounts receivable and inventory, fixed assets, other, and unsecured or blanket lien), each interacted with COVID. Interest rate controls include interest rate spread, and an indicator variable for whether interest rate spread is reported (this only occurs for non-fixed rate credit lines with > 0% drawdown); in this specification, all fixed rate loans are dropped. Robust standard errors are clustered at the NAICS three-digit level in parentheses; *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Economic message. The crisis liquidity gap between small and large firms is not a tiny residual fact. It is economically large, statistically strong, and partly explained by the restrictive contract

terms documented earlier.

5.5 Tables 12–14 and Figures 3–4: similar exposure, different drawdowns

Table 12
Revenue growth by industry exposure and size.

Dep. var.:	2020 Q1-Q3 YoY revenue growth in size-industry cell	
	Mean (1)	Median (2)
0-250 (in mil.)	-6.81* (3.08)	-5.10** (3.17)
1000+	-12.11** (2.92)	-12.24** (3.00)
Exposure < 0-250	-13.22** (3.34)	-14.14** (3.33)
Exposure > 1000+	-16.28** (3.19)	-16.19** (3.28)
$P(\beta_{\text{small}0-250} = \beta_{\text{small}1000+})$	0.51	0.47
$P(\beta_{\text{medium}0-250} = \beta_{\text{medium}1000+})$	0.50	0.66
Observations	129	129

Notes: The table reports regressions of mean (column 1) or median (column 2) revenue growth through 2020Q3 relative to 2019Q1-2019Q3 within a size-industry cell. Size class is based on assets (in millions) in 2019Q4. Exposure is the three-digit NAICS code industry-level growth in employment between 2019Q2 and 2020Q2 less the average Q2-to-Q2 growth in the industry between 2015 and 2019, standardized to have unit variance across industries.

Table 13
Drawdowns by firm size and exposure to COVID-19 shock: abnormal three-digit industry decline in employment.

Dependent variable	Drawdown rate (in ppt)					
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure < COVID	3.1	-0.4	0.8	0.9	0.9	0.6
Exposure > 50-250 (in mil.) < COVID	(2.3)	3.5**	2.4**	2.2**	2.1**	0.9*
Exposure > 250-1000 < COVID		1.3	0.8	0.8	0.7	0.5
		4.4*	3.3*	3.3*	3.4*	1.2
Exposure > 1000-5000 < COVID		2.1	1.0	1.4	1.4	1.0
		7.2**	6.1**	6.1**	6.1**	2.6**
Exposure > 5000 < COVID		2.2	1.7	1.5	1.4	1.0
		9.5**	8.2**	8.2**	7.8**	4.0**
0-250 (in mil.) < COVID		4.6**	3.9**	3.7**	3.6**	0.8*
		10.3	8.0	8.0	8.0	1.8
250-1000 < COVID		11.3**	9.5**	9.2**	9.2**	4.0**
		1.2	1.0	0.9	0.9	0.6
1000-5000 < COVID		15.4**	13.2**	12.9**	12.3**	5.6**
		1.2	1.1	1.0	1.0	0.7
5000+ < COVID		18.0**	15.1**	14.7**	14.1**	5.5**
		2.6	1.8	1.9	2.0	1.2
Loan FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank-Time FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Time FE	No	No	No	No	No	No
Franchise	No	No	No	No	No	No
Rating-Time FE	No	No	No	No	No	No
Loan Terms	No	No	No	No	No	No
No of Firms	60,117	60,117	60,117	60,117	60,117	60,117
N	756,529	756,529	756,529	756,527	685,096	497,754
R ²	0.81	0.81	0.81	0.81	0.86	0.86

Notes: Results from estimating a model of the following type: $\Delta \text{Drawdown}_{i,t} = \alpha_0 + \alpha_1 \sum_{j=1}^J \beta_j \text{Exposure}_{i,t} + \alpha_2 \text{Exposure}_{i,t} \times \text{Size}_{i,t} + \alpha_3 \text{COVID}_{i,t} + \alpha_4 \text{COVID}_{i,t} \times \text{Exposure}_{i,t} + \alpha_5 \text{COVID}_{i,t} \times \text{Size}_{i,t} + \alpha_6 \text{COVID}_{i,t} \times \text{Exposure}_{i,t} \times \text{Size}_{i,t} + \alpha_7 \text{COVID}_{i,t} \times \text{Exposure}_{i,t} \times \text{Size}_{i,t} \times \text{COVID}_{i,t}$, where $\Delta \text{Drawdown}_{i,t}$ is the rate of decline in employment, COVID is an indicator variable for 2020Q2 and 2020Q3, and Exposure is the three-digit NAICS code industry-level growth in employment between 2019Q2 and 2020Q2 less the average Q2-to-Q2 growth in the industry between 2015 and 2019. We restrict the sample to outstanding loans from 2017Q4 onwards. Rating-time fixed effects are categorical variables for 10 internal loan rating categories. Firm financial controls are lagged debtors, cash and receivables, operating income plus depreciation and amortization, assets, year-over-year sales growth, and operating income/asset. Each interaction with COVID. Loan term controls are six maturity categories (less than 6 months, 6-12 months, 1-2 years, 2-4 years, and more than 4 years), six collateral classes (real estate, marketable securities, accounts receivable and inventory, fixed assets, other, and unsecured or blanket lien), 5 categories of borrower size in COVID (< 250, 250-499, 500-999, 1000-4999, and 5000+), and interest rate spreads, each in levels and interacted with COVID. Robust standard errors are clustered at the three-digit NAICS industry level in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 14
Instrumenting industry exposure with physical proximity needs.

Dependent variable	$\Delta \text{Drawdown}_{i,t} \times \text{expos}_{i,t}$ (in ppt)						
	OLS		2SLS				
Firm size	All		<\$50	\$50-\$250	\$250-\$1000	\$1000-\$5000	>\$5000
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Exposure	3.8**	3.6	-1.8	1.9	4.2*	2.3**	12.8**
	(1.4)	(2.4)	(2.4)	(2.8)	(2.5)	(2.1)	(4.8)
F-Statistic (N=7)	17.475	16.912	16.901	14.247	15.984	8.745	8.945
No of Firms	43,806	43,806	29,184	7195	3488	2493	1336
N	67,981	67,981	33,040	9812	7452	8722	8905

Notes: This table shows results from estimating a model of the following type: $\Delta \text{Drawdown}_{i,t} \times \text{expos}_{i,t} = \alpha_0 + \alpha_1 \text{Exposure}_{i,t} + \alpha_2 \text{COVID}_{i,t} + \alpha_3 \text{COVID}_{i,t} \times \text{Exposure}_{i,t} + \alpha_4 \text{COVID}_{i,t} \times \text{Size}_{i,t} + \alpha_5 \text{COVID}_{i,t} \times \text{Exposure}_{i,t} \times \text{Size}_{i,t} + \alpha_6 \text{COVID}_{i,t} \times \text{Exposure}_{i,t} \times \text{Size}_{i,t} \times \text{COVID}_{i,t}$, where $\Delta \text{Drawdown}_{i,t} \times \text{expos}_{i,t}$ is the difference in firm i 's and Exposure, is the three-digit NAICS code industry-level growth in employment between 2019Q2 and 2020Q2 less the average Q2-to-Q2 growth in the industry between 2015 and 2019. In column (2) through (7), we instrument Exposure, with the exposure to the COVID survey question: "How physically close to other people are you when you perform your normal job?" aggregated to the industry-level. Effective F-statistic reported according to Staiger and Poirer (2015). Standard errors are clustered by three-digit NAICS code.

Read:

- Table 12 validates the exposure measure. A one-standard-deviation increase in industry exposure predicts about a 15 percentage point decline in revenue growth for both small and large firms.
- Figure 3 shows the central scatterplot result: exposure predicts drawdowns strongly for large firms, but almost not at all for SMEs.
- Table 13 formalizes this in the triple-difference regressions. The marginal effect of exposure is essentially zero for the smallest firms but rises monotonically with size. In the simple specification it reaches about 9.5 percentage points for the largest firms, and remains sizable after rich controls.
- Figure 4 shows the timing: there are no obvious pre-trends, and the exposure effect for large firms jumps sharply in 2020Q1.
- Table 14 instruments exposure using physical proximity. The IV results keep the same pattern: negligible response for firms below \$50 million, but very large response for the biggest firms.

Economic message. This is the most persuasive evidence in the paper. Small and large firms were hit by similar industry-level shocks, but only large firms translated those shocks into credit-line borrowing. That is very hard to explain without contract heterogeneity.

5.6 Table 15 and Figure 5: what PPP did

Read:

- The paper identifies 51,713 current Y-14 borrowers as PPP recipients, and 97% of matched PPP loans with nonmissing assets go to SMEs.
- Table 15 shows that for small SME recipients, non-PPP borrowing falls from \$103.0B in 2020Q1 to \$79.5B in 2020Q2, while PPP funds received equal \$32.8B. This implies a repayment ratio of about 71.7%.

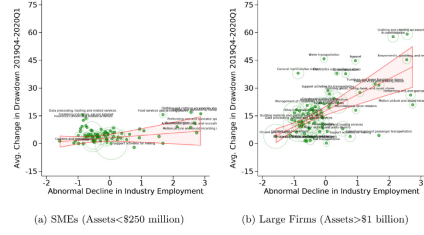


Fig. 3. Exposure to COVID-shock and credit line drawdowns for SMEs and large firms. Abnormal employment decline is the three-digit NAICS code industry-level growth in employment between 2019Q2 and 2020Q2 less the average Q2-to-Q2 growth in the industry between 2015 and 2019. We add linear fits with industries weighted by number of firms per industry. Data are restricted to industries with at least ten firms per firm size category. Perimeter of hollow circles indicate relative industry size by number of firms reporting in the Y14 within the respective size class.

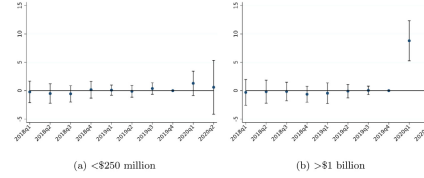


Fig. 4. Dynamics of credit line drawdowns for SMEs and large firms during the COVID recession. The figure plots the sequence of coefficients (β_t) obtained from estimating $\text{Drawdown}_{i,t} = \alpha_i + \beta_t \times \text{Exposure}_i + \epsilon_{i,t}$, where $\text{Drawdown}_{i,t}$ is the ratio of utilized to committed credit and Exposure_i is the three-digit NAICS code industry-level growth in employment between 2019Q2 and 2020Q2 less the average Q2-to-Q2 growth in the industry between 2015 and 2019. Coefficients are normalized to 2019Q4 and 95% confidence bands.

Table 15

Aggregate drawdowns for PPP participants by firm size, 2019Q4–2020Q2.

	Non-PPP credit outstanding (\$Bil)			PPP amount (\$Bil)	Repayment ratio (%)	N
	2019Q4	2020Q1	2020Q2			
Firm assets (\$mil)						
Not classified	11.4	11.5	10.8	3.6	19.5	6857
0–50	101.6	103.0	79.5	32.8	71.7	38,508
50–250	68.9	69.7	57.1	11.5	109.0	5055
250–1000	22.0	23.7	20.4	1.6	201.2	935
1000–5000	11.2	16.6	12.3	0.3	1431.3	248
5000–	7.8	12.5	9.6	0.1	2268.8	110
Sum	222.7	237.0	189.7	50.0	94.7	51,713

Notes: The table reports the total dollar amount (in \$B) of non-PPP credit outstanding (left-most three columns), total PPP funds received, and the ratio of the change in credit outstanding between 2020Q1 and 2020Q2 to PPP funds received for the PPP recipients identified in the Y-14.

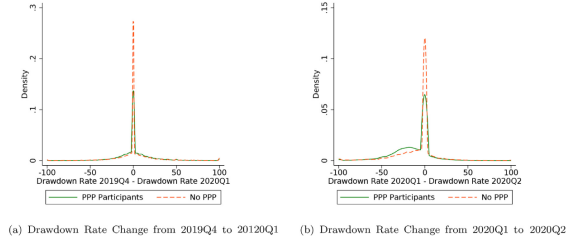


Fig. 5. Kernel density of drawdowns at small SMEs.

- Pooling across all matched PPP borrowers, the repayment ratio is about 94.7%.
- Figure 5 shows that PPP recipients and nonrecipients look similar in 2020Q1, but PPP recipients are much more likely to repay non-PPP debt in 2020Q2.

Economic message. PPP looks less like a pure extra credit expansion and more like a public substitute for private liquidity that small firms were unable to obtain through ordinary credit-line drawdowns when the shock hit.

6 Short summary

The paper shows that small firms and large firms hold very different bank loan contracts, and those differences matter most in a crisis. Small firms are much more likely to have short-maturity or demandable credit lines, to post collateral, to use lines heavily in normal times, and yet still to pay higher spreads. The authors interpret this bundle of terms as giving lenders more discretion. When COVID hit, large firms drew heavily on pre-existing committed lines, especially in exposed industries, while small firms did not, even after extensive controls for borrower condition, bank condition, location, industry, and bond-market access. Loan terms explain an important part of this gap. PPP then effectively substituted for the private liquidity that many small firms could not access on their own.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that access to bank liquidity depends crucially on contract design, and that small firms tend to borrow under contracts that leave much more discretion to lenders. In normal times that discretion can support lending to opaque firms. In crisis times it means small firms cannot rely on their posted credit limits the same way large firms can.

7.2 Economic intuition

The main intuition is that commitment and discretion solve different problems. Commitment gives insurance against liquidity shocks. Discretion protects the lender when borrower quality is uncertain. Small firms, being more opaque and volatile, sort into contracts that make discretion valuable ex ante. But once a large shock arrives, those same contracts make liquidity much less dependable.

7.3 Takeaway

The paper's big lesson is that "available credit" is not a pure quantity concept. It is a contingent contractual object. For small firms in particular, undrawn commitments can drastically overstate actual access to liquidity when the economy turns down.